

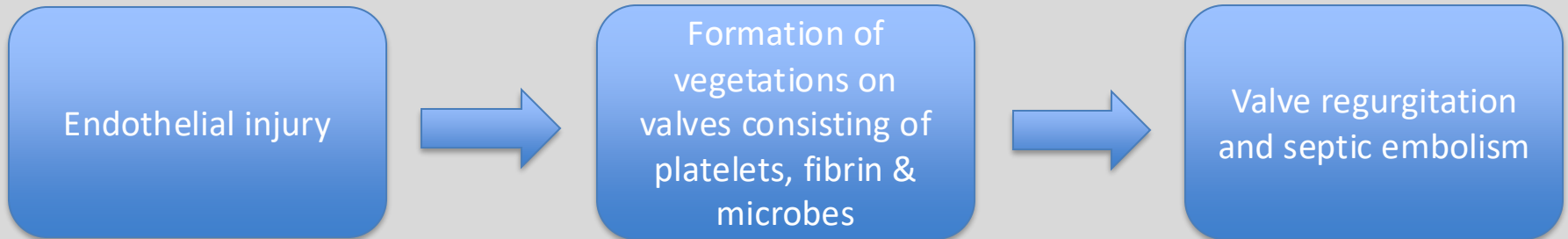
Session Objectives

1. Describe the etiologies associated with endocarditis, myocarditis, and pericarditis.
2. Explain the clinical presentation of each condition
3. Explain how each condition can be accurately diagnosed.
4. Describe other related clinical forms of inflammatory heart disease, including marantic endocarditis, tamponade, and constrictive pericarditis.

Risk Factors

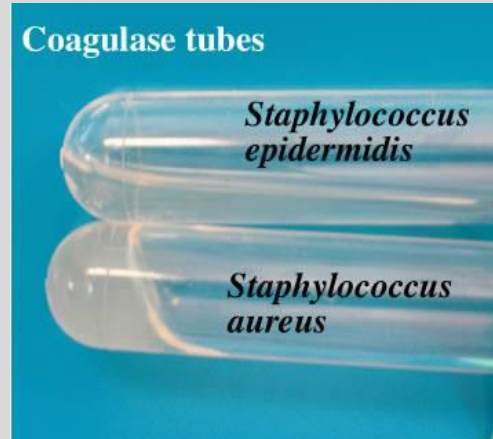
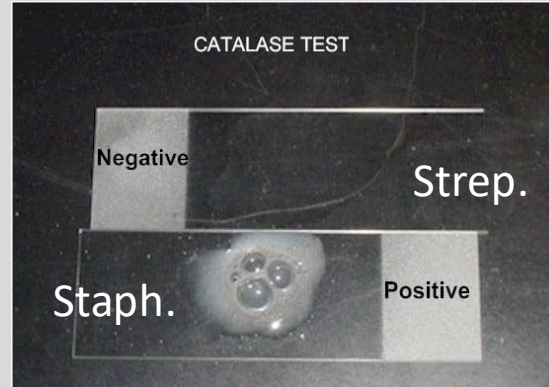
- Age >60
- Injection drug use
- Poor dentition/dental infection
- History of infective endocarditis
- Structural heart disease (valvular, congenital)
- Chronic hemodialysis
- Procedures (intravascular device, cardiac implantable electronic device, cardiac bypass surgery)

Pathogenesis



Staphylococcus

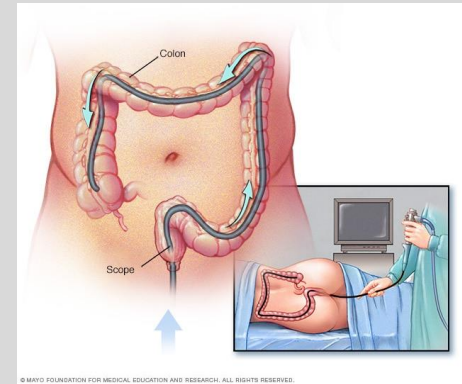
- *Staphylococcus aureus*
 - Gram +, Catalase +, Coagulase +
 - Causes **acute** endocarditis
 - Tends to infect *normal* (native) heart valves
- *Staphylococcus epidermidis*
 - Gram +, Catalase +, Coagulase –
 - Low virulence bacteria found in normal skin flora
 - Associated with prosthetic valves



Streptococcus

- Viridans group *Streptococcus* → *S. sanguinis*, *S. mitis*, *S. mutans*
 - Group of Gram + cocci, Catalase –
 - Found in normal mouth flora; IE may occur after a dental procedure
 - Typically cause subacute (over several weeks) endocarditis that often affect *damaged* heart valves.
- *Streptococcus gallolyticus (bovis)*
 - Gram + cocci found in normal gut bacteria
 - Associated with colon cancer

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Diagnosis of Infective Endocarditis based on Modified Duke criteria:

Patient needs to have one of the following:

- 2 major criteria
- 1 major and 3 minor criteria
- 5 minor criteria

MAJOR:

- Positive blood cultures
- Vegetations on echocardiogram

MINOR:

- Fever ($\geq 38.0^{\circ}\text{C}$ or 100.4°F)
- Risk factors
- Roth spots, Osler nodes, Janeway lesions, splinter hemorrhages

Differential Diagnosis

Bacteremia + valvular vegetations = endocarditis

However, in the presence of only one finding, consider other differentials:

- Bacteremia but NO evidence of valvular vegetations → infection from an intravascular catheter, cardiac device, or prosthetic joint, etc.
- Valvular vegetations or arterial emboli but NO bacteremia → culture-negative endocarditis (Libman-Sacks), antiphospholipid syndrome

Etiologies

Inflammation of the myocardium can develop from various causes:

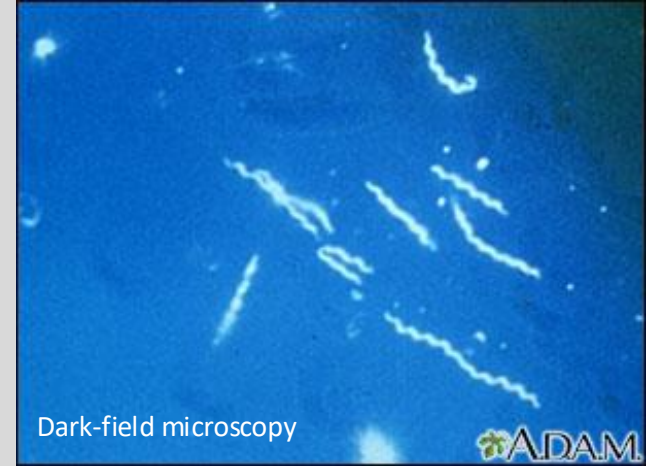
- 50% of cases are idiopathic
- Infection – viral, bacterial, protozoal
- Toxins – ethanol, illicit drug use, chemotherapeutics, environmental toxins
- Immunologic – allergic responses or systemic disorders
- Physical agents – electric shock, radiation, trauma

Lyme Carditis

Caused by the spirochete *Borrelia burgdorferi*

Stage 1: early localized infection

Erythema migrans from tick bite



Stage 2: early disseminated infection
(days to months later)

Transient colonization in bloodstream
may cause secondary skin lesions, AV
block, myocarditis

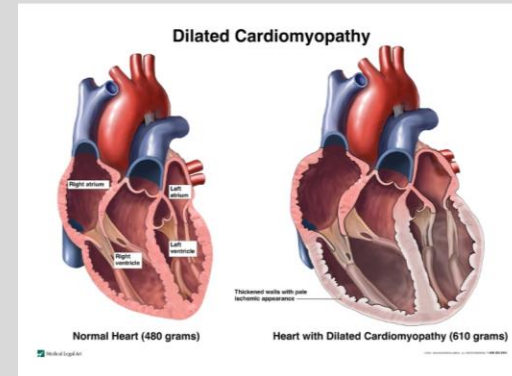


Stage 3: late stage infection (months to years later)

Chronic infection in multiple tissue sites can cause
Lyme arthritis, neurological issues

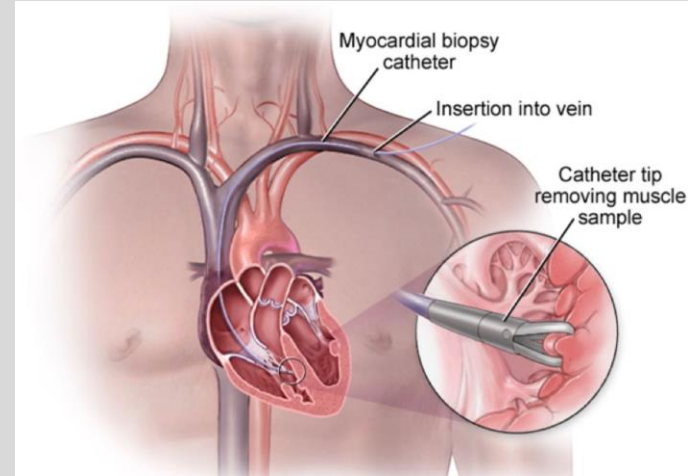
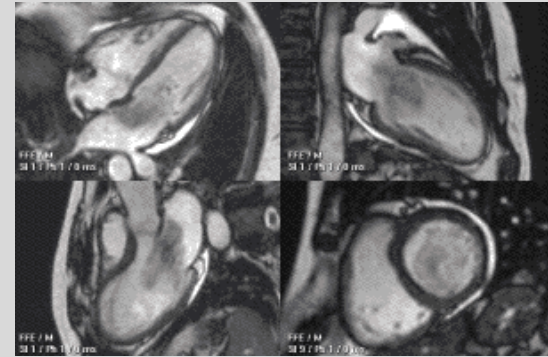
Clinical Presentation

- Patient *may be asymptomatic*, or symptoms may be similar to the flu:
 - Fever, chest pain and other constitutional symptoms (e.g., headache, fatigue, muscle aches)
- Myocarditis is rarely fulminant
- Complication: Chronic myocarditis can cause persistent ventricular dysfunction, which can lead to **heart failure (dilated cardiomyopathy)**



Workup

- Initial evaluation: ECG, troponins, CBC, inflammatory markers (CRP, ESR), echocardiogram
- Gold standard noninvasive test = **cardiac magnetic resonance imaging (CMR)**
- Gold standard diagnostic test = **Endomyocardial biopsy (EMB)**
 - Indicated in select cases, such as unexplained fulminant heart failure, life-threatening arrhythmias, etc.
 - You cannot make a definitive diagnosis of myocarditis without EMB that meets Dallas criteria
- Treatments and interventions should be individualized based on clinical context and comorbidities



Clinical Presentation & Diagnosis

- Symptoms:
 - Chest pain – sharp, worsens with inhalation and lying supine, better when sitting up/leaning forward
 - Fever
- **Pericardial friction rub** – a scratchy sound heard in systole and diastole
- ECG changes - ST elevation, PR depression
- Pericarditis can involve **pericardial effusion** – excess fluid (serous, blood, pus, or lymphatic fluid) in the pericardial space



Cardiomegaly due to a massive pericardial effusion. At least 200 mL of fluid enlarges the silhouette.

Cardiac Tamponade

Tamponade = high pericardial pressure compressing the heart

- Fluid around the heart = pericardial effusion → When the effusion compresses the heart → cardiac tamponade
- Decreased cardiac output
- Clinical presentation: Beck's triad = hypotension, jugular venous distention (JVD), muffled heart sounds

Constrictive Pericarditis

Chronic inflammation and fibrous constriction of the pericardium

- Pericardial fibrosis and calcification restricts space for expansion → decreased ventricular filling
- Clinical presentation: dyspnea, Kussmaul sign (increased jugular venous pressure), pericardial knock (high-pitched sound in early diastole)
- Most common cause resource-limited countries = **Tuberculosis**

Summary

- **Endocarditis:** Infection of endocardium/valves; often *S. aureus*, *Strep viridans*, enterococci. Risk factors: IV drug use, prosthetic valves, structural heart disease. Related: Marantic endocarditis
 - Clinical: Fever, new murmur, embolic/immune lesions (Roth spots, Osler nodes, Janeway lesions)
 - Dx: Duke's criteria, blood cultures, TTE/TEE
- **Myocarditis:** Inflammation of myocardium; often viral (can be bacterial or protozoal)
 - Clinical: Variable; can mimic MI; can lead to cardiomyopathy
 - Dx: Imaging $\pm$ biopsy (Dallas criteria)
- **Pericarditis:** Inflammation of pericardium; often viral (can be bacterial or fungal)
 - Clinical: Sharp chest pain, better leaning forward; friction rub; ECG change
 - Dx: Clinical + ECG, imaging to assess effusion
 - Complication: tamponade, chronic constriction